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# MAST20030: Differential Equations
# Tutorial Solutions: Week 7

**Solution to Question 1.**

(a) Firstly, all eigenfunctions satisfy the boundary conditions, i.e.,

$$\begin{aligned} y_0(-\pi) = y_0(\pi) &= 1/\sqrt{2}, \\ y_0'(-\pi) = y_0'(\pi) &= 0, \\ y_n^{(1)}(-\pi) = y_n^{(1)}(\pi) &= (-1)^n, \\ y_n'^{(1)}(-\pi) = y_n'^{(1)}(\pi) &= 0, \\ y_n^{(2)}(-\pi) = y_n^{(2)}(\pi) &= 0, \\ y_n'^{(2)}(-\pi) = y_n'^{(2)}(\pi) &= n(-1)^n. \end{aligned}$$

Next we check the differential operator,

$$-\frac{\mathrm{d}^2}{\mathrm{d}x^2} y_0 = 0$$

therefore $\lambda = 0$ is the eigenvalue for eigenfunction $y_0$. Similarly

$$-\frac{\mathrm{d}^2}{\mathrm{d}x^2} y_n^{(2)} = n^2 \cos(nx) = n^2 y_n^{(1)} \qquad \text{and} \qquad -\frac{\mathrm{d}^2}{\mathrm{d}x^2} y_n^{(2)} = n^2 \sin(nx) = n^2 y_n^{(2)}$$

and so $\lambda_n = n^2$ are the eigenvalues for $y_n^{(1)}$ and $y_n^{(2)}$.

(b) We first calculate the inner product $\langle y_m^{(2)}, y_n^{(2)} \rangle$ where $m \neq n$:

$$\langle y_m^{(2)}, y_n^{(2)} \rangle = \alpha \int_{-\pi}^{\pi} \sin(mx)\sin(nx)\,\mathrm{d}x = \frac{\alpha}{2} \int_{-\pi}^{\pi} \Big[\cos(m-n)x - \cos(m+n)x\,\mathrm{d}x\Big] = 0$$

because $m - n$ and $m + n$ are always non-zero integers. Now

$$\langle y_n^{(2)}, y_n^{(2)} \rangle = \alpha \int_{-\pi}^{\pi} \sin^2(nx)\,\mathrm{d}x = \frac{\alpha}{2} \int_{-\pi}^{\pi} \Big[1 - \cos(2nx)\Big]\mathrm{d}x = \frac{\alpha}{2}\,[x]_{-\pi}^{\pi} = \alpha\pi$$

and so the eigenfunctions are orthogonal. To make the eigenfunctions orthonormal we set $\alpha = 1/\pi$.

**Solution to Question 2.**

(a) We can write $\sin^2 x = \frac{1}{2}\big(1 - \cos(2x)\big)$, so we should suspect that the Fourier series of $f$ is just $\frac{1}{2} - \frac{1}{2}\cos(2x)$. To check this, note that $f$ is an even function, so the coefficient of each $\sin(nx)$ vanishes. The Fourier series therefore takes the form

$$\mathcal{F}\{f\} = F(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos(nx),$$

where

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,\mathrm{d}x = \frac{1}{\pi}\int_{-\pi}^{\pi} \left[\frac{1}{2} - \frac{1}{2}\cos(2x)\right]\mathrm{d}x = \frac{1}{\pi}\cdot 2\pi \cdot \frac{1}{2} = 1$$

(because $\cos(2x)$ is orthogonal to 1)

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,\mathrm{d}x = \frac{1}{\pi}\int_{-\pi}^{\pi}\left[\frac{1}{2} - \frac{1}{2}\cos(2x)\right]\cos(nx)\,\mathrm{d}x = -\frac{1}{2\pi}\int_{-\pi}^{\pi}\cos(2x)\cos(nx)\,\mathrm{d}x$$

(because $\cos(nx)$ is orthogonal to 1, for $n \neq 0$)

$$= -\frac{1}{2\pi}\delta_{n,2}\int_{-\pi}^{\pi}\cos^2(2x)\,\mathrm{d}x$$

(because $\cos(nx)$ is orthogonal to $\cos(2x)$, if $n \neq 2$)

$$= -\frac{1}{2\pi}\delta_{n,2}\int_{-\pi}^{\pi}\left[\frac{1}{2} + \frac{1}{2}\cos(4x)\right]\mathrm{d}x = -\frac{1}{2}\delta_{n,2}$$

(because $\cos(4x)$ is orthogonal to 1). So, indeed, the Fourier series of $f$ is $\mathcal{F}\{f\} = F(x) = \frac{1}{2} - \frac{1}{2}\cos(2x)$.

(b) Again, $g$ is an even function, so the coefficient of each $\sin(\pi nx)$ vanishes. We therefore compute

$$a_0 = \int_{-1}^{1} g(x)\,\mathrm{d}x = \int_{-1}^{1} x\,\mathrm{d}x = \frac{1}{2}x^2\Big|_{-1}^{1} = 1,$$

$$a_n = \int_0^1 g(x)\cos(\pi nx)\,\mathrm{d}x = 2\int_0^1 x\cos(\pi nx)\,\mathrm{d}x = 2\,\frac{x\sin(\pi nx)}{\pi n}\Big|_0^1 - 2\int_0^1 \frac{\sin(\pi nx)}{\pi n}\,\mathrm{d}x$$

$$= 2\,\frac{\cos(\pi nx)}{\pi^2 n^2}\Big|_0^1 = \frac{2((-1)^n - 1)}{\pi^2 n^2} = \begin{cases} -\dfrac{4}{\pi^2 n^2} & \text{if } n > 0 \text{ is odd,} \\ 0 & \text{if } n > 0 \text{ is even.} \end{cases}$$

The Fourier series of $g$ is therefore

$$G(x) = \mathcal{F}\{g\}(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n\cos(\pi nx) = \frac{1}{2} - \frac{4}{\pi^2}\sum_{m=1}^{\infty}\frac{\cos(\pi(2m-1)x)}{(2m-1)^2}.$$

As this converges to the periodic extension of $g$ and $g(0) = 0$, we deduce that

$$\frac{1}{2} - \frac{4}{\pi^2}\sum_{m=1}^{\infty}\frac{1}{(2m-1)^2} = 0 \quad \Rightarrow \quad \sum_{m=1}^{\infty}\frac{1}{(2m-1)^2} = \frac{\pi^2}{8}.$$

For the last result, note that

$$\sum_{n=1}^{\infty}\frac{1}{n^2} = \sum_{m=1}^{\infty}\frac{1}{(2m)^2} + \sum_{m=1}^{\infty}\frac{1}{(2m-1)^2} = \frac{1}{4}\sum_{n=1}^{\infty}\frac{1}{n^2} + \sum_{m=1}^{\infty}\frac{1}{(2m-1)^2} \;\Rightarrow\; \sum_{n=1}^{\infty}\frac{1}{n^2} = \frac{4}{3}\sum_{m=1}^{\infty}\frac{1}{(2m-1)^2} = \frac{\pi^2}{6}.$$

**Solution to Question 3.**

(a) The function looks like in the figure below.

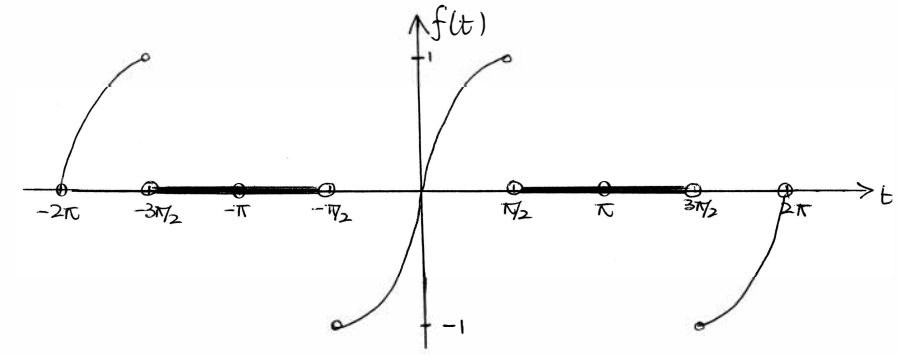
(b) By the 'pointwise convergence of a Fourier series' theorem, the Fourier series will converge to the midpoint at the discontinuity at $t = \pi/2$, i.e., the Fourier series will converge to $1/2$.

(c) Both the constant and cos terms will be zero because the integrands for the coefficients is odd. The coefficients for the sin terms are

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t) = \frac{1}{\pi}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\sin t\sin(nt)\,\mathrm{d}t = \frac{1}{2\pi}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\Big[\cos(n-1)t - \cos(n+1)t\Big]\mathrm{d}t.$$

Now for $n = 1$,

$$b_1 = \frac{1}{2\pi}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\Big[1 - \cos 2t\Big]\mathrm{d}t = \frac{1}{2\pi}\left[t + \frac{1}{2}\sin(2t)\right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \frac{1}{2}.$$




For $n \neq 1$,

$$b_n = \frac{1}{2\pi}\left[\frac{1}{n-1}\sin(n-1)t - \frac{1}{n+1}\sin(n+1)t\right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$$
$$= \frac{1}{\pi}\left(\frac{1}{n-1}\sin(n-1)\frac{\pi}{2} - \frac{1}{n+1}\sin(n+1)\frac{\pi}{2}\right).$$

Now $b_{2n-1} = 0$ and

$$b_{2n} = \frac{1}{\pi}\left(\frac{1}{2n-1}(-1)^{n+1} - \frac{1}{2n+1}(-1)^n\right) = \frac{(-1)^{n+1}}{\pi}\left(\frac{1}{2n-1} + \frac{1}{2n+1}\right) = \frac{4n(-1)^{n+1}}{(4n^2-1)\pi}$$

The Fourier series is therefore given by

$$F(x) = \mathcal{F}\{f\} = \frac{1}{2}\sin t + \sum_{n=1}^{\infty}\frac{4n(-1)^{n+1}}{(4n^2-1)\pi}\sin(2nt)$$

(d) At $t = \dfrac{\pi}{2}$, $\sin 2nt = 0$ and $\sin t = 1$ and so $(\mathcal{F}f)\left(\dfrac{\pi}{2}\right) = \dfrac{1}{2}$.

(e) The Fourier coefficients decay as $\dfrac{1}{n}$ for large $n$, and so the convergence will be slow. Because the periodic extension of the function contains discontinuities at $\dfrac{\pi}{2} + m\pi$ for $m \in \mathbb{Z}$ the Fourier series does not converge uniformly and will exhibit the Gibbs phenomenon.